



ClearClause



ClearClause: Privacy Policy Seeker and Summarizer

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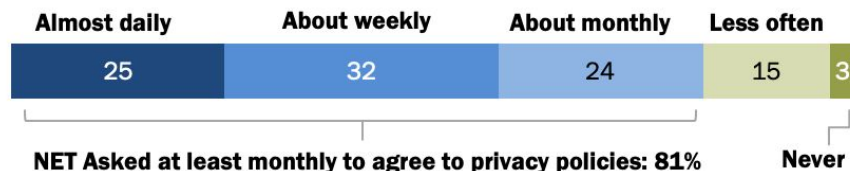
Problem:

People don't read privacy policies.

- 23% of people often or always read them
- Widespread lack of understanding between consumers and corporations
- Roos (2017) pointed out that they are created for legal grounds for data collection and not to inform users.
- 45 million Americans are functionally illiterate

A majority of Americans are asked to agree to privacy policies at least monthly, including a quarter who say this happens daily

% of U.S. adults who say they are asked to agree to the terms and conditions of a company's privacy policy ...



Note: Those who did not give an answer are not shown.

Source: Survey conducted June 3-17, 2019.

"Americans and Privacy: Concerned, Confused and Feeling Lack of Control Over Their Personal Information"

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Background



What are Privacy Policies?

Legal documents that outline how a website collects, uses, stores, and shares user data.

- Privacy Policies aren't actually required by any federal laws
- Regional legislation like the CPRA (California) and GDPR (EU) force companies into creating them

Other Solutions: Government and Corporate Action

A few companies such as Twitter/X and Facebook have tried to make their policies simpler but they still seem to fall short.

- Twitter Made theirs into a game in 2022
 - “No one has time for that” - Geoffrey Fowler (New York Times)

The wait for federal privacy legislation could be long

- California won't settle for anything less than their CCPA/ CPRA



Other Solutions: PrivaSeer

- Privacy policy search engine
 - Can find policy text or URLs
- Utilizes Common Crawl and Free Company Dataset



Cons

- Can be hard to sort through all returned links
- Only 1.4 million privacy policies indexed

Other Solutions: ToS;DR

- Project Started in 2012 in Berlin, Germany
- Created to fix the “biggest lie on the web”
- Analyzes each aspect of a privacy policy (Positive, Negative, Neutral)
 - Entire policy receives a grade from A to E
- Has a browser extension that shows grade and description of policy



Cons

- Summaries are user generated
 - Sites will not have a summary if they are less popular

Why is mine Better?

Real time

Summaries generated at the user's request

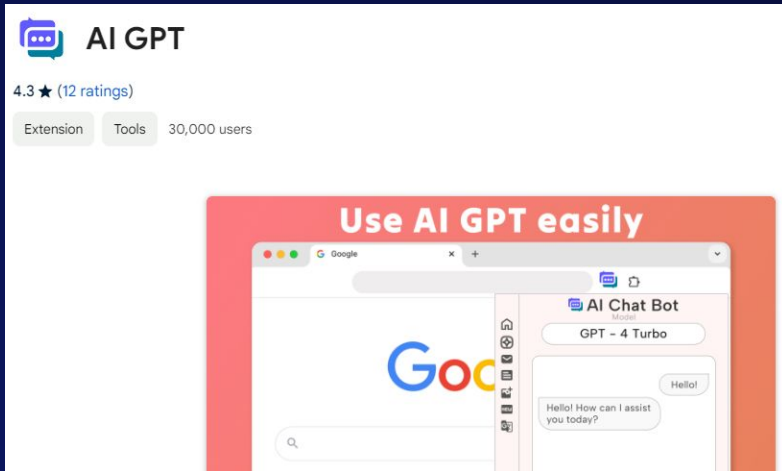
Ease of Use

The only work required on the user's part is to open the extension and press a button.

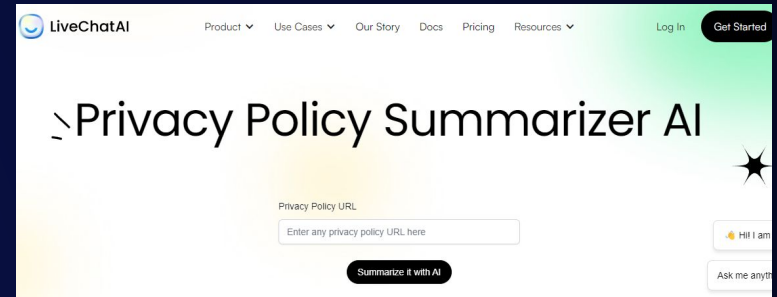


Novelty

While both AI browser extensions and Privacy Policy summarizers are quite common, the combination of the two has not been done before.



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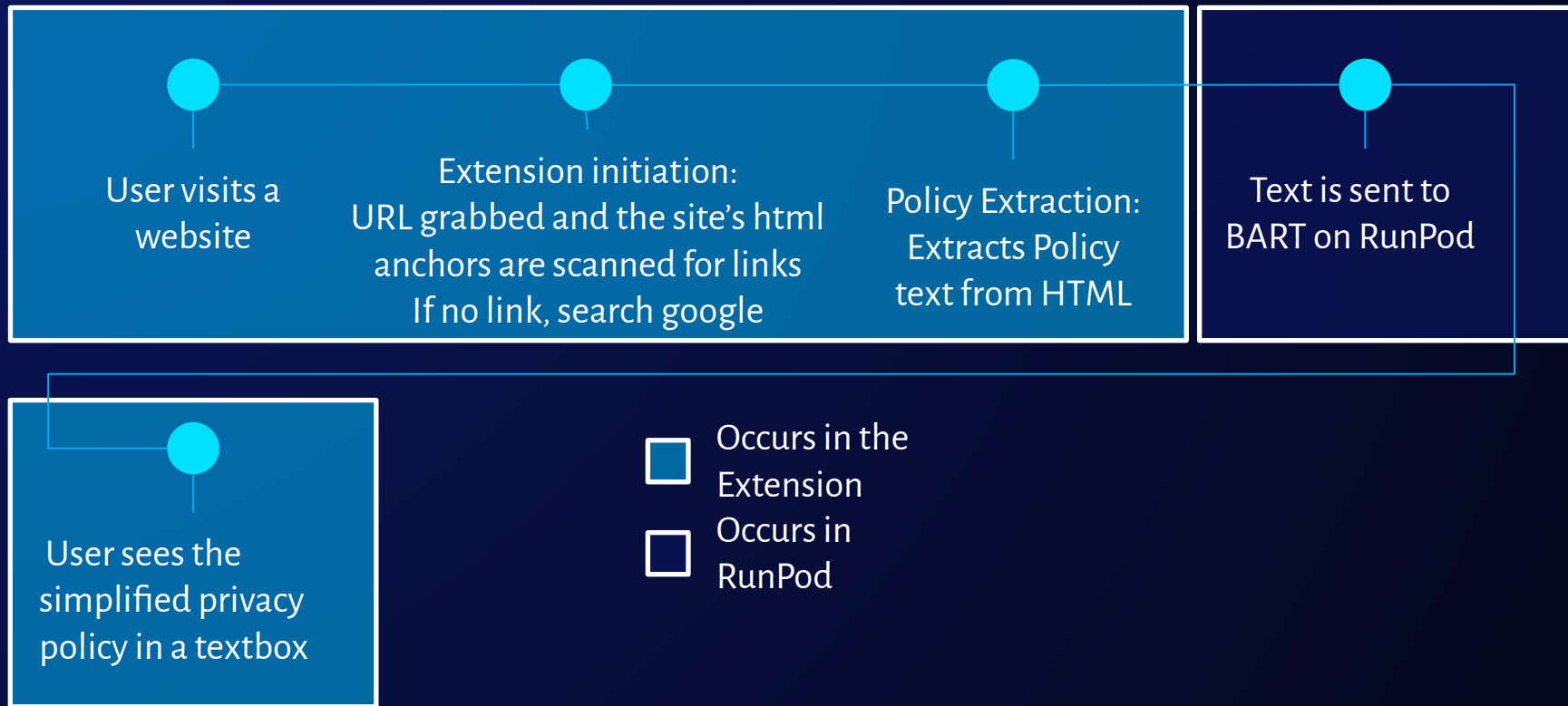


Impact

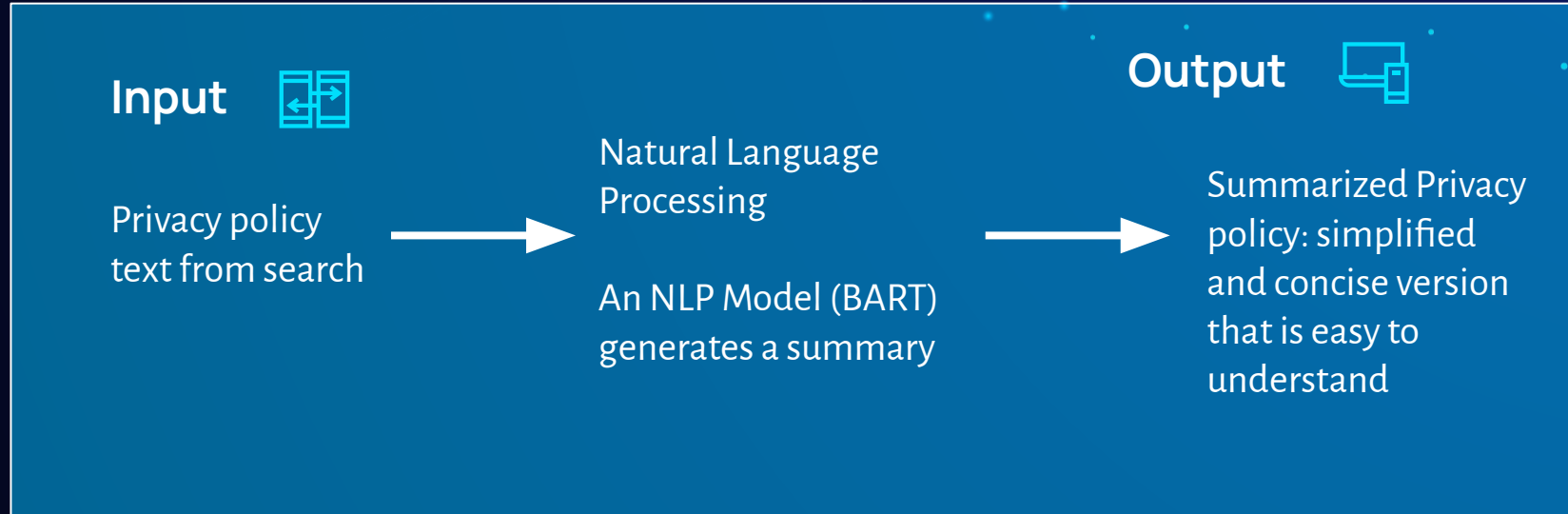
- Allow users to make more informed decisions when it comes to the sites that they use.
- Foster trust between companies and users
 - Consumers will be more aware of the ways that a company acts with their personal data.



Method: Systems Architecture



Method



Tech Stack:

- JavaScript
- HTML
- NLP model (BART)
- Runpod
 - Docker
 - Python

Methods: Summarization

Text is preprocessed

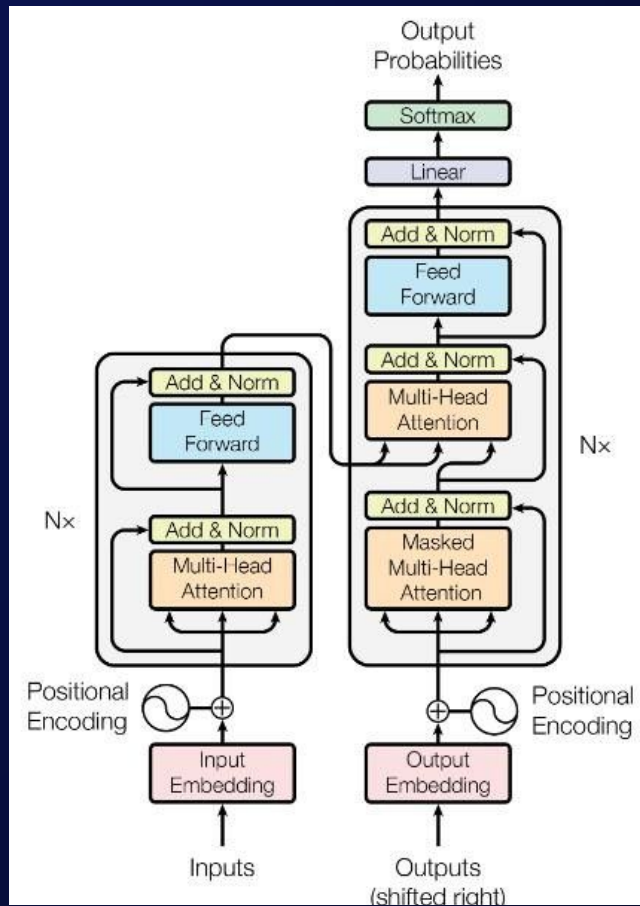
- Tokenization
- Special tokens added (e.g. start of sequence)
- Text is converted to standard format (lowercase, no punctuation)



Text is passed through a Facebook's BART (NLP) abstractive summarization algorithm

- Context present in input is very important
- word-by-word summary is generated

Methods: BART model (NLP)



Methods:Tokenization

```
print(vocab)
```

```
['<|endoftext|>', ',', '.', 'C', 'F', 'H', 'T', 'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'k', 'l', 'm', 'n', 'o',  
'p', 'r', 's', 't', 'u', 'v', 'w', 'y', 'z', 'Ġ', 'Ġt', 'is', 'er', 'Ġa', 'Ġto', 'en', 'Th', 'This', 'ou', 'se',  
'Ġtok', 'Ġtoken', 'nd', 'Ġis', 'Ġth', 'Ġthe', 'in', 'Ġab', 'Ġtokeni']
```

```
tokenize("This is not a token.")
```

```
['This', 'Ġis', 'Ġ', 'n', 'o', 't', 'Ġa', 'Ġtoken', '.']
```

Methods: Vectorization

Embedding Matrix:

"un"	[0.1, 0.2, ..., 0.9]
"believ"	[0.5, -0.3, ..., 0.8]
"able"	[0.2, 0.5, ..., 0.8]
"re"	[0.6, -0.2, ..., -0.7]

Input: "unbelievable"

Tokens: ["un", "believ", "able"]

"un" $\rightarrow$ [0.1, 0.2, ..., 0.9]

"believ" $\rightarrow$ [0.5, -0.3, ..., 0.8]

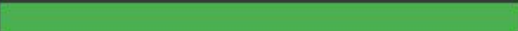
Methods: BART model (NLP)

```
from transformers import pipeline
```

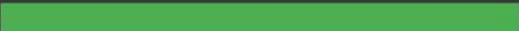
```
summarizer = pipeline("summarization", model="facebook/bart-large-cnn")
```

```
... /usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/).
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
```

```
warnings.warn(
```

```
config.json: 100%  1.58k/1.58k [00:00<00:00, 112kB/s]
```

```
model.safetensors: 100%  1.63G/1.63G [00:06<00:00, 253MB/s]
```

```
generation_config.json: 100%  363/363 [00:00<00:00, 38.1kB/s]
```

```
vocab.json: 100%  899k/899k [00:00<00:00, 4.50MB/s]
```

```
merges.txt: 100%  456k/456k [00:00<00:00, 7.29MB/s]
```

```
tokenizer.json: 100%  1.36M/1.36M [00:00<00:00, 6.90MB/s]
```

```
Device set to use cuda:0
```

```
ngrok public URL: NgrokTunnel: "https://9d15-35-227-152-161.ngrok-free.app" -> "http://localhost:5000"
```

Pseudocode (RunPod)

In RunPod, after server is started:

From Hugging Face: Load summarizer pipeline

```
summarize_large_text(text):  
    chunk_list = chunk_text()  
    Summary = ""  
    For chunk in chunk_list  
        Add bullet point + summarize(chunk) to summary
```

When app is run:

```
Prompt = Take input from request event  
Return summarize(prompt)
```

```
summarize(text):  
    Summarize using summarizer pipeline
```

Pseudocode

In extension (JavaScript):

FindPolicy()

- try to find policy link on active tab

- if fail

 - return error message

- if successful

 - send content to be summarized

 - (also report errors during summarization)

findPrivacyPolicyLink() – actual function that looks at links/anchors and whether they contain the word “privacy”; also conducts google searches

summarizeText()

- send post request to summarization API

- handle any errors

Methods: Extension Code

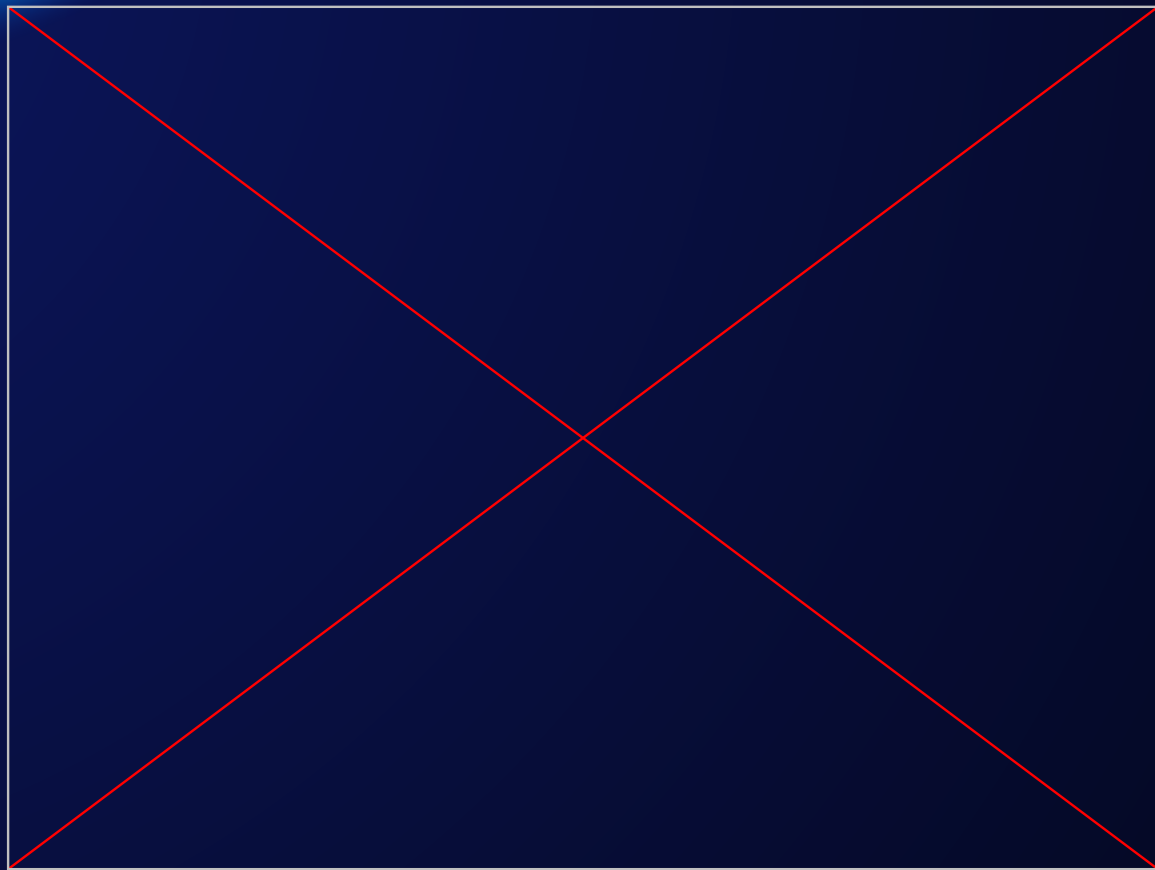
```
const checkCurrentPageForPolicy = () => {
  const bodyText = document.body.innerText || '';
  const title = document.title || '';
  const url = window.location.href || '';
  const metaTags = Array.from(document.getElementsByTagName('meta')).map(meta => meta.content || '').join(' ');

  // Combine the title, URL, and meta tags into one searchable string
  const searchableText = `${title} ${url} ${metaTags}`;

  // Check for keywords in title, URL, or meta tags
  if (/((privacy policy|your privacy|data usage|data protection))/i.test(searchableText)) {
    return bodyText;
  }
  return null;
};
```

```
async function summarizeText(text) {
  const response = await fetch("https://2d11-34-139-114-49.ngrok-free.app/summarize", {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
    },
    body: JSON.stringify({ text: text }),
  });
}
```

Demo



Results

- Summary ~40% the length of the original text (moderate length original, 2000 words)
- Summaries on moderate length policies generated within 1-2 minutes usually
- Summaries work on many popular websites, though hidden elements may cause issues at times.



Limitations



Language

Privacy Policies will only be found if they are in English



Accuracy

There's no way to summarize text without losing some nuance and detail



Policy Availability

Srinath et al. (2023) Found that only 34% of websites had an accessible privacy policy link

Conclusion and Future Work

My extension will continue to provide accurate and useful summaries of privacy policies to users in order to help them make more informed decisions regarding their data.

In the future I hope to...

- Expand to more languages
- Help users find contact information for websites without a privacy policy
- Improve summaries

References

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Q&A